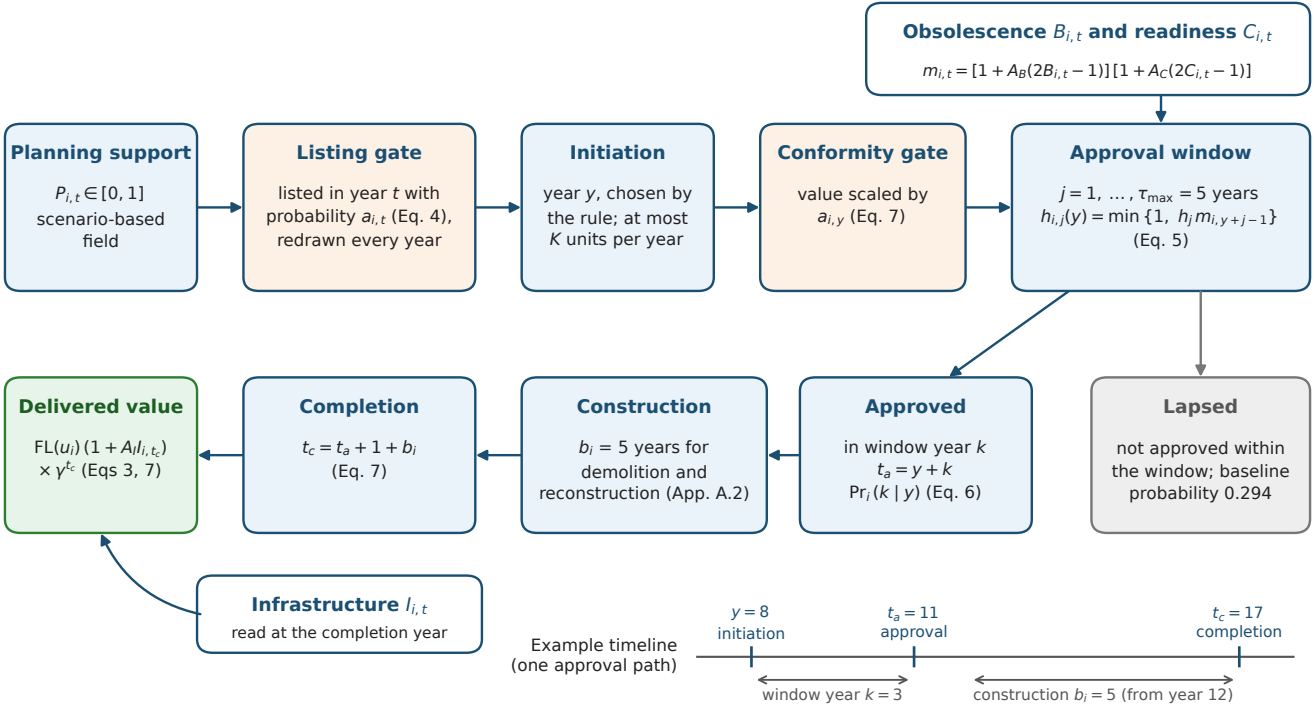

Obsolescence $B_{i,t}$ and readiness $C_{i,t}$
$m_{i,t} = [1 + A_B(2B_{i,t} - 1)][1 + A_C(2C_{i,t} - 1)]$
Planning support
$P_{i,t} \in [0, 1]$
scenario-based field
Listing gate
listed in year t with probability $a_{i,t}$ (Eq. 4), redrawn every year
Initiation
year y, chosen by the rule; at most K units per year
Conformity gate
value scaled by $a_{i,y}$ (Eq. 7)
Approval window
$j = 1, \ldots, \tau_{\max} = 5$ years
$h_{i,j}(y) = \min\{1, h_j m_{i,y+j-1}\}$
(Eq. 5)
Approved
in window year k
$t_a = y + k$
$\Pr_i(k \mid y)$ (Eq. 6)
Lapsed
not approved within the window; baseline probability 0.294
Construction
$b_i = 5$ years for demolition and reconstruction (App. A.2)
Completion
$t_c = t_a + 1 + b_i$
(Eq. 7)
Delivered value
$\mathrm{FL}(u_i)\,(1 + A_I I_{i,t_c}) \times \gamma^{t_c}$ (Eqs 3, 7)
Infrastructure $I_{i,t}$
read at the completion year
Example timeline (one approval path)
$y = 8$ initiation
$t_a = 11$ approval
$t_c = 17$ completion
window year $k = 3$
construction $b_i = 5$ (from year 12)